

Basic theory of a standing wave optical cavity in vacuum

Alexei Gurchenko

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1 Reflected, transmitted and intracavity fields

Let us start by calculating the reflected, transmitted and intracavity fields. Figure 1 depicts the schematic of the electromagnetic wave traveling inside the optical cavity. We assume that the mirrors 1 and 2 have corresponding amplitude transmission and reflection coefficients r_1 , t_1 and r_2 , t_2 . The intracavity medium is lossless, however there is a loss of amplitude due to scattering at the mirror surfaces: t_a at Mirror 1 and t_b at Mirror 2. This model is the closest approximation of a two-mirror cavity inside ultra-high vacuum environment.

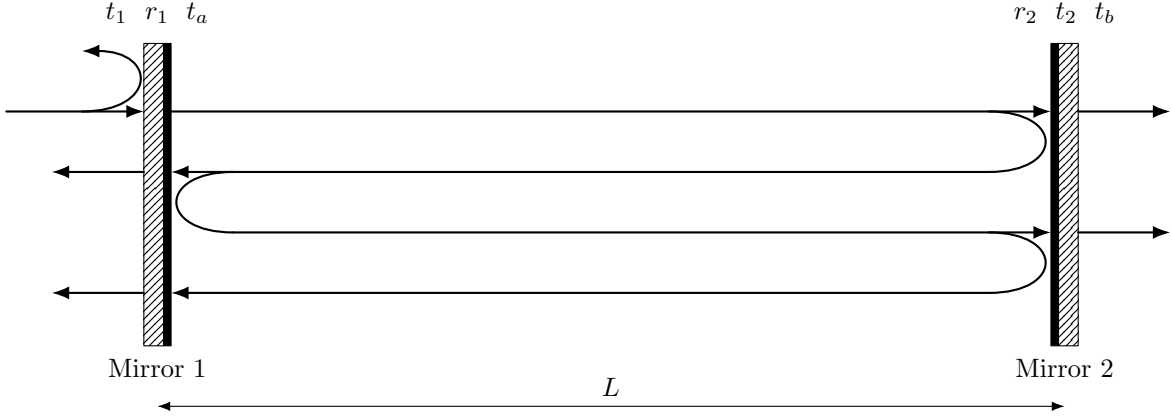


Figure 1: Optical cavity schematic

1.1 Reflection and transmission at an interface

The first question we need to answer before moving on to calculating reflected field, is the following. If light is shined to an interface from one side and the interface has reflected coefficient r and transmitted coefficient t , what would be the reflected and transmitted coefficients r' and t' if the light is shined from another side? This elegant derivation [1] of the relation between the field transmission and reflection coefficients was made by the British physicist Sir George Gabriel Stokes. He assumed that there was no dissipation at the interface and therefore the process obeyed time-reversal invariance. The three waves are shown on the left in the figure. The time-reversed case appears in the center plot. When we reverse time, we need to include three additional waves: the reflection from the bottom wave, and the transmission of both the bottom and top waves. All of the waves are shown in the plot on the right. In order for the right-hand plot to be equivalent to the time-reversed center plot, the two waves in the upper left must sum to E and the two waves in the lower left must sum to zero. Thus,

$$E = E t t' + E r r' \text{ and } 0 = E r t + E r' t$$

References

- [1] Warren Nagourney. *Quantum Electronics for Atomic Physics and Telecommunication*. 2014. DOI: 10.1093/acprof:oso/9780199665488.001.0001.